

Quantum Field Theory

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Preface

This is my notes on Quantum Field Theory. The main reference is *Field Quantization* by Walter Greiner and *Introduction to Quantum Field Theory* by Horatiu Nastase. Other references include *Quantum Theory of Fields* by Steven Weinberg, *Quantum Field Theory of Particles and Strings* by Brian Hatfield, *Modern Quantum Field Theory: A Concise Introduction* by Tom Banks, and *Quantum Field Theory and the Standard Model* by Matthew D. Schwarz.

This note does not attempt to provide a comprehensive treatment of relativistic quantum mechanics. Instead, it contains only a concise overview of selected topics, namely classical field theory and the representations of the Poincaré group.

Chapter 1

Scalar Field

1.1 Classical Scalar Field

1.2 Quantization of Free Scalar Field

1.3 Schrödinger/Heisenberg/Interaction Picture

1.4 Green Function and Propagator

1.5 Correlation Function

Chapter 2

Representations of Poincare Group

2.1 Poincare Group

2.2 Lie Algebra of Poincare Group

2.3 Representations of Poincare Group

Chapter 3

Spinor Field

3.1 Classical Spinor Field

3.2 Gamma Matrices

3.3 Canonical Quantization of Free Spinor Field

3.4 Green Function and Propagator

3.5 Correlation Function

Chapter 4

Vector Field

4.1 Classical Proca Field and Classical Maxwell Field

4.2 Canonical Quantization of Proca Field

4.3 Canonical Quantization of Maxwell Field

4.3.1 Gauge Quantization

4.3.2 Covariant Quantization

4.4 Green Function and Propagator

4.5 Correlation Function

Chapter 5

Exact Solutions

5.1 Free Fields

5.1.1 Free Scalar Field

5.1.2 Free Spinor Field

5.1.3 Free Vector Field

5.2 Sinh-Gordon Model

5.3 Massive Thirring Model

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Quantum Properties

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Density Operator Formalism

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Non-Relativistic Quantum Field Theory

9.1 Non-Relativistic Scalar Field

9.2 Representation of Galilean Group

9.3 Non-Relativistic Spinor Field

9.4 Non-Relativistic Vector Field

9.5 Low-Energy Limit of Relativistic Quantum Field Theory

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Symmetry

10.1 Symmetry and Conservation

10.2 Symmetry and Degeneracy

10.3 Parity Reversal

10.4 Time Reversal

10.5 Charge Conjugation

10.6 CPT Theorem

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Interacting Field Theory

11.1 List of Interacting Field Theories

11.2 Interaction Picture and Dyson's Formula

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Renormalization of QED

12.1 IR Divergence in QED

12.2 UV Divergence in QED

12.3 Regularization

12.4 Renormalization of QED

12.5 Renormalization Group

Chapter 13

Non-Abelian Gauge Theory

13.1 $U(1)$ Gauge Theory

13.2 Classical Yang–Mills Theory

13.3 Canonical Quantization of Yang–Mills Theory

Chapter 14

QCD

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Renormalization of QCD

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Spontaneous Symmetry Breaking

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Electroweak Theory

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Neutrino

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Anomalies

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Grand Unified Theory

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Finite-Temperature Quantum Field Theory

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Supersymmetry

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Supersymmetric Quantum Field Theory

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Spontaneous Supersymmetric Breaking

Chapter 25

QFT in Curved Spacetime

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Stress-Tensor Renormalization

Chapter 27

Quantum Black Hole